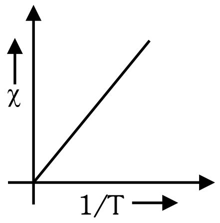
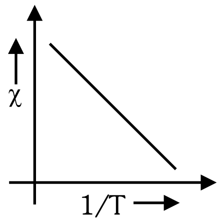
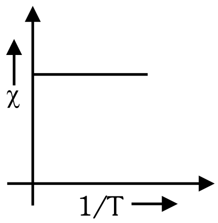
NEET(UG) – 2023(Manipur)

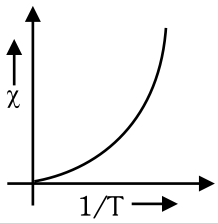
Section-A (Physics)

1. The variation of susceptibility (χ) with absolute temperature (T) for a paramagnetic material is represented as :

(1) 

(2) 

(3) 

(4) 

2. A bullet of mass m hits a block of mass M elastically. The transfer of energy is the maximum, when :

(1) M = m
(2) M = 2m
(3) M << m
(4) M >> m

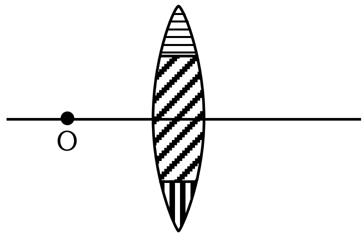
3. The ground state energy of hydrogen atom is –13.6 eV. The energy needed to ionize hydrogen atom from its second excited state will be :

(1) 13.6 eV
(2) 6.8 eV
(3) 1.51 eV
(4) 3.4 eV

4. The escape velocity of a body on the earth surface is 11.2 km/s. If the same body is projected upward with velocity 22.4 km/s, the velocity of this body at infinite distance from the centre of the earth will be:

(1) $11.2\sqrt{2}$ km/s
(2) Zero
(3) 11.2 km/s
(4) $11.2\sqrt{3}$ km/s

5. A lens is made up of 3 different transparent media as shown in figure. A point object O is placed on its axis beyond $2f$. How many real images will be obtained on the other side ?



(1) 2
(2) 1
(3) No image will be formed
(4) 3

6. The diameter of a spherical bob, when measured with vernier callipers yielded the following values : 3.33 cm, 3.32 cm, 3.34 cm, 3.33 cm and 3.32 cm. The mean diameter to appropriate significant figures is :

(1) 3.328 cm
(2) 3.3 cm
(3) 3.33 cm
(4) 3.32 cm

7. On the basis of electrical conductivity, which one of the following material has the smallest resistivity?

(1) Germanium
(2) Silver
(3) Glass
(4) Silicon

8. The mechanical quantity, which has dimensions of reciprocal of mass (M^{-1}) is :

(1) angular momentum
(2) coefficient of thermal conductivity
(3) torque
(4) gravitational constant

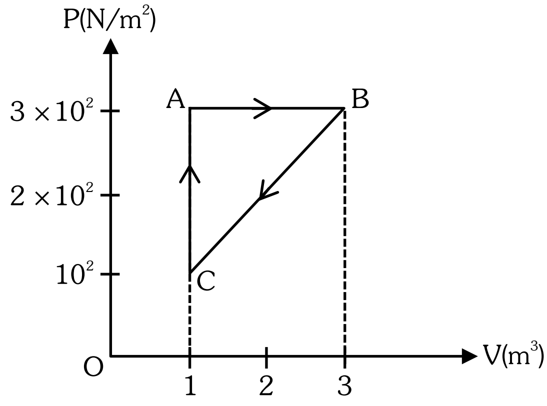
9. The position of a particle is given by

$$\vec{r}(t) = 4t\,\hat{i} + 2t^2\,\hat{j} + 5\,\hat{k}$$

where t is in seconds and r in metre. Find the magnitude and direction of velocity v(t), at t = 1s, with respect to x-axis

(1) $4\sqrt{2}$ ms^{-1}, 45°
(2) $4\sqrt{2}$ ms^{-1}, 60°
(3) $3\sqrt{2}$ ms^{-1}, 30°
(4) $3\sqrt{2}$ ms^{-1}, 45°

10. For the given cycle, the work done during isobaric process is :



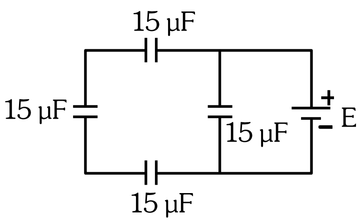
(1) 200 J

(2) Zero

(3) 400 J

(4) 600 J

11. The equivalent capacitance of the arrangement shown in figure is :



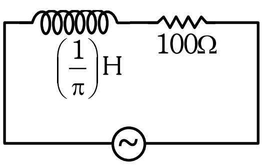
(1) 30 μF

(2) 15 μF

(3) 25 μF

(4) 20 μF

12. An ac source is connected in the given circuit. The value of ϕ will be :



$V = 220 \sin(100\pi t + \phi)$ volt

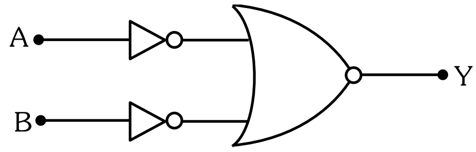
(1) 60°

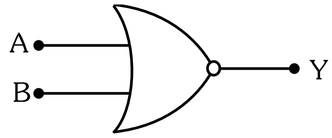
(2) 90°

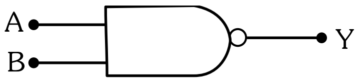
(3) 30°

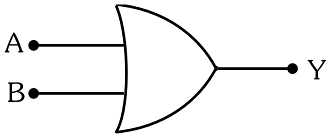
(4) 45°

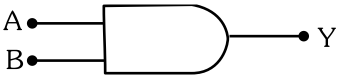
13. The given circuit is equivalent to :



(1) 

(2) 

(3) 

(4) 

14. A particle moves with a velocity $\left(5\hat{i}-3\hat{j}+6\hat{k}\right)$ m s^{-1} horizontally under the action of constant force $\left(10\hat{i}+10\hat{j}+20\hat{k}\right)$N. The instantaneous power supplied to the particle is :

(1) 200 W (2) Zero

(3) 100 W (4) 140 W

15. A certain wire A has resistance 81 Ω. The resistance of another wire B of same material and equal length but of diameter thrice the diameter of A will be :

(1) 81 Ω (2) 9 Ω

(3) 729 Ω (4) 243 Ω

16. $\in_0$ and μ_0 are the electric permittivity and magnetic permeability of free space respectively. If the corresponding quantities of a medium are $2\in_0$ and $1.5\mu_0$ respectively, the refractive index of the medium will nearly be :

(1) $\sqrt{2}$ (2) $\sqrt{3}$

(3) 3 (4) 2

17. The amount of elastic potential energy per unit volume (in SI unit) of a steel wire of length 100 cm to stretch it by 1 mm is (if Young's modulus of the wire = 2.0×10^{11} Nm^{-2}) :

(1) 10^{11} (2) 10^{17}

(3) 10^{7} (4) 10^{5}

18. The 4th overtone of a closed organ pipe is same as that of 3rd overtone of an open pipe. The ratio of the length of the closed pipe to the length of the open pipe is :

(1) 8 : 9 (2) 9 : 7
(3) 9 : 8 (4) 7 : 9

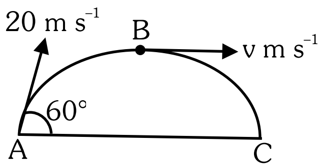
19. A long straight wire of length 2 m and mass 250 g is suspended horizontally in a uniform horizontal magnetic field of 0.7 T. The amount of current flowing through the wire will be ($g = 9.8\ ms^{-2}$) :

(1) 2.45 A
(2) 2.25 A
(3) 2.75 A
(4) 1.75 A

20. According to Gauss law of electrostatics, electric flux through a closed surface depends on :

(1) the area of the surface
(2) the quantity of charges enclosed by the surface
(3) the shape of the surface
(4) the volume enclosed by the surface

21. A ball is projected from point A with velocity 20 m s^{-1} at an angle 60° to the horizontal direction. At the highest point B of the path (as shown in figure), the velocity v m s^{-1} of the ball will be:



(1) 20
(2) $10\sqrt{3}$
(3) Zero
(4) 10

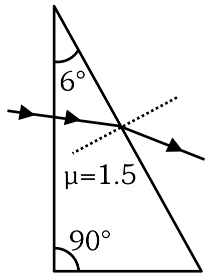
22. Which of the following statement is not true ?

(1) Coefficient of viscosity is a scalar quantity
(2) Surface tension is a scalar quantity
(3) Pressure is a vector quantity
(4) Relative density is a scalar quantity

23. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron:

(1) will turn towards right of direction of motion
(2) will turn towards left of direction of motion
(3) speed will decrease
(4) speed will increase

24. A horizontal ray of light is incident on the right angled prism with prism angle 6°. If the refractive index of the material of the prism is 1.5, then the angle of emergence will be:

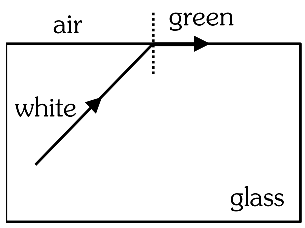


(1) 9° (2) 10° (3) 4° (4) 6°

25. A p-type extrinsic semiconductor is obtained when Germanium is doped with:

(1) Antimony
(2) Phosphorous
(3) Arsenic
(4) Boron

26.



Which set of colours will come out in air for a situation shown in figure?

(1) Yellow, Orange and Red
(2) All
(3) Orange, Red and Violet
(4) Blue, Green and Yellow

27. If Z_1 and Z_2 are the impedances of the given circuits (a) and (b) as shown in figures, then choose the **correct** option

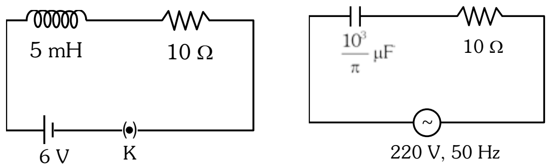


figure (a) figure (b)

(1) $Z_1 < Z_2$ (2) $Z_1 + Z_2 = 20\ \Omega$

(3) $Z_1 = Z_2$ (4) $Z_1 > Z_2$

28. The wavelength of Lyman series of hydrogen atom appears in:

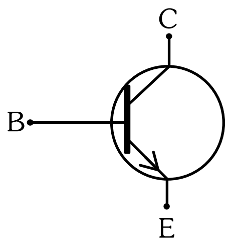
(1) visible region

(2) far infrared region

(3) ultraviolet region

(4) infrared region

29.



The above figure shows the circuit symbol of a transistor. Select the **correct** statements given below:

(A) The transistor has two segments of p-type semiconductor separated by a segment of n-type semiconductor.

(B) The emitter is of moderate size and heavily doped.

(C) The central segment is thin and lightly doped.

(D) The emitter base junction is reverse biased in common emitter amplifier circuit.

(1) (C) and (D) (2) (A) and (D)

(3) (A) and (B) (4) (B) and (C)

30. The de Broglie wavelength associated with an electron, accelerated by a potential difference of 81 V is given by:

(1) 13.6 nm (2) 136 nm

(3) 1.36 nm (4) 0.136 nm

31. The maximum power is dissipated for an ac in a/an:

(1) resistive circuit (2) LC circuit

(3) inductive circuit (4) capacitive circuit

32. The maximum kinetic energy of the emitted photoelectrons in photoelectric effect is independent of :

(1) work function of material

(2) intensity of incident radiation

(3) frequency of incident radiation

(4) wavelength of incident radiation

33. Two particles A and B initially at rest, move towards each other under mutual force of attraction. At an instance when the speed of A is v and speed of B is 3v, the speed of centre of mass is :

(1) 2v (2) zero

(3) v (4) 4v

34. A charge Q μC is placed at the centre of a cube. The flux coming out from any one of its faces will be (in SI unit) :

(1) $\frac{Q}{\epsilon_0} \times 10^{-6}$ (2) $\frac{2Q}{3\epsilon_0} \times 10^{-3}$

(3) $\frac{Q}{6\epsilon_0} \times 10^{-3}$ (4) $\frac{Q}{6\epsilon_0} \times 10^{-6}$

35. The viscous drag acting on a metal sphere of diameter 1 mm, falling through a fluid of viscosity 0.8 Pa s with a velocity of 2 m s^{-1} is equal to :

(1) 15×10^{-3} N

(2) 30×10^{-3} N

(3) 1.5×10^{-3} N

(4) 20×10^{-3} N

Section-B (Physics)

36. If R is the radius of the earth and g is the acceleration due to gravity on the earth surface. Then the mean density of the earth will be :

(1) $\frac{\pi RG}{12g}$ (2) $\frac{3\pi R}{4gG}$ (3) $\frac{3g}{4\pi RG}$ (4) $\frac{4\pi G}{3gR}$

37. A copper wire of radius 1 mm contains 10^{22} free electrons per cubic metre. The drift velocity for free electrons when 10 A current flows through the wire will be (Given, charge on electron = 1.6×10^{-19} C) :

(1) $\frac{6.25 \times 10^4}{\pi} \, m s^{-1}$ (2) $\frac{6.25}{\pi} \times 10^3 m s^{-1}$

(3) $\frac{6.25}{\pi} m s^{-1}$ (4) $\frac{6.25 \times 10^5}{\pi} m s^{-1}$

38. An object is mounted on a wall. Its image of equal size is to be obtained on a parallel wall with the help of a convex lens placed between these walls. The lens is kept at distance x in front of the second wall. The required focal length of the lens will be :

(1) less than $\frac{x}{4}$

(2) more than $\frac{x}{4}$ but less than $\frac{x}{2}$

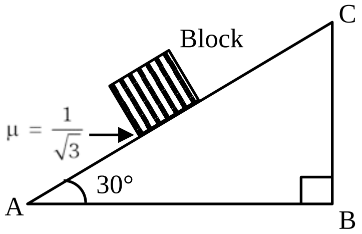
(3) $\frac{x}{2}$

(4) $\frac{x}{4}$

39. If a conducting sphere of radius R is charged. Then the electric field at a distance r(r > R) from the centre of the sphere would be, (V = potential on the surface of the sphere)

(1) $\frac{rV}{R^2}$ (2) $\frac{R^2V}{r^3}$ (3) $\frac{RV}{r^2}$ (4) $\frac{V}{r}$

40. A block of mass 2 kg is placed on inclined rough surface AC (as shown in figure) of coefficient of friction μ. If g = 10 m s^{-2}, the net force (in N) on the block will be :



(1) $10\sqrt{3}$ (2) zero (3) 10 (4) 2

41. A container of volume 200 cm^3 contains 0.2 mole of hydrogen gas and 0.3 mole of argon gas. The pressure of the system at temperature 200 K (R = 8.3 JK^{-1} mol^{-1}) will be :-

(1) 6.15×10^5 Pa

(2) 6.15×10^4 Pa

(3) 4.15×10^5 Pa

(4) 4.15×10^6 Pa

42. To produce an instantaneous displacement current of 2 mA in the space between the parallel plates of a capacitor of capacitance 4 μF, the rate of change of applied variable potential difference $\left(\frac{dV}{dt}\right)$ must be :-

(1) 800 V/s (2) 500 V/s

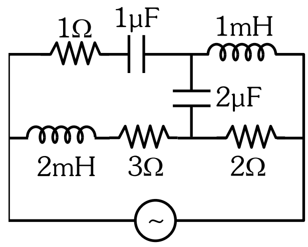
(3) 200 V/s (4) 400 V/s

43. An emf is generated by an ac generator having 100 turn coil, of loop area 1 m^2. The coil rotates at a speed of one revolution per second and placed in a uniform magnetic field of 0.05 T perpendicular to the axis of rotation of the coil. The maximum value of emf is :-

(1) 3.14 V (2) 31.4 V

(3) 62.8 V (4) 6.28 V

44. For very high frequencies, the effective impedance of the circuit (shown in the figure) will be :-



(1) 4 Ω (2) 6 Ω

(3) 1 Ω (4) 3 Ω

45. A constant torque of 100 N-m turns a wheel of moment of inertia 300 kg-m^2 about an axis passing through its centre. Starting from rest, its angular velocity after 3s is :-

(1) 1 rad/s (2) 5 rad/s

(3) 10 rad/s (4) 15 rad/s

46. The emf of a cell having internal resistance 1Ω is balanced against a length of 330 cm on a potentiometer wire. When an external resistance of 2Ω is connected across the cell, the balancing length will be :

(1) 220 cm (2) 330 cm
(3) 115 cm (4) 332 cm

47. A 1 kg object strikes a wall with velocity 1 m s^{-1} at an angle of 60° with the wall and reflects at the same angle. If it remains in contact with wall for 0.1 s, then the force exerted on the wall is :-

(1) $30\sqrt{3}$ N (2) Zero
(3) $10\sqrt{3}$ N (4) $20\sqrt{3}$ N

48. The angular momentum of an electron moving in an orbit of hydrogen atom is $1.5\left(\frac{h}{\pi}\right)$. The energy in the same orbit is nearly.

(1) –1.5 eV (2) –1.6 eV
(3) –1.3 eV (4) –1.4 eV

49. A particle is executing uniform circular motion with velocity $\vec{v}$ and acceleration $\vec{a}$. Which of the following is true ?

(1) $\vec{v}$ is a constant; $\vec{a}$ is not a constant.
(2) $\vec{v}$ is not a constant; $\vec{a}$ is not a constant.
(3) $\vec{v}$ is a constant; $\vec{a}$ is a constant.
(4) $\vec{v}$ is not a constant; $\vec{a}$ is a constant.

50. A simple pendulum oscillating in air has a period of $\sqrt{3}$ s. If it is completely immersed in non-viscous liquid, having density $\left(\frac{1}{4}\right)^{th}$ of the material of the bob, the new period will be :-

(1) $2\sqrt{3}$ s (2) $\frac{2}{\sqrt{3}}$ s (3) 2s (4) $\frac{\sqrt{3}}{2}$ s

Section-A (Chemistry)

51. **Incorrect** set of quantum numbers from the following is :

(1) n=4, ℓ = 3, m_ℓ = –3, –2, –1, 0, +1, +2, +3, m_s = –1/2
(2) n=5, ℓ = 2, m_ℓ = –2, –1, +1, +2, m_s = +1/2
(3) n=4, ℓ = 2, m_ℓ = –2, –1, 0, +1, +2, m_s = –1/2
(4) n= 5, ℓ = 3, m_ℓ = –3, –2, –1, 0, +1, +2, +3, m_s = +1/2

52. Given below are two statements : one is labelled as **Assertion (A)** & the other is labelled as **Reason (R)**.

Assertion (A) :

Ionisation enthalpy increases along each series of the transition elements from left to right. However, small variations occur.

Reason (R) :

There is corresponding increase in nuclear charge which accompanies the filling of electrons in the inner d-orbitals.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **(A)** is correct but **(R)** is not correct.
(2) **(A)** is not correct but **(R)** is correct.
(3) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.
(4) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**.

53. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) :

Lithium and beryllium unlike their other respective group members form compounds with pronounced ionic character.

Reason (R) :

Lithium and Magnesium have similar properties due to diagonal relationship.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **(A)** is true but **(R)** is false.
(2) **(A)** is false but **(R)** is true.
(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(4) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**.

54. For a weak acid HA, the percentage of dissociation is nearly 1% at equilibrium. If the concentration of acid is 0.1 mol L^{-1}, then the **correct** option for its K_a at the same temperature is :

(1) 1×10^{-4}
(2) 1×10^{-6}
(3) 1×10^{-5}
(4) 1×10^{-3}

55. The density of 1 M solution of a compound 'X' is 1.25 g mL^{-1}. The **correct** option for the molality of solution is (Molar mass of compound X = 85 g) :

(1) 0.705 m
(2) 1.208 m
(3) 1.165 m
(4) 0.858 m

56. Consider the given reaction :

$$CH_3COCH_3 \xrightarrow{\text{dil}\,Ba(OH)_2} \text{"X"}$$

The functional groups present in compound "X" are:

(1) ketone and double bond
(2) double bond and aldehyde
(3) alcohol and aldehyde
(4) alcohol and ketone

57. The $E^{\ominus}$ values for

$Al^{+}/Al = +0.55$ V and $Tl^{+}/Tl = -0.34$ V

$Al^{3+}/Al = -1.66$ V and $Tl^{3+}/Tl = +1.26$ V

Identify the incorrect statement

(1) Al is more electropositive than Tl
(2) Tl^{3+} is a good reducing agent than Tl^{1+}
(3) Al^{+} is unstable in solution
(4) Tl can be easily oxidised to Tl^{+} than Tl^{3+}

58. The correct order of dipole moments for molecules NH_3, H_2S, CH_4 and HF, is:

(1) $CH_4 > H_2S > NH_3 > HF$
(2) $H_2S > NH_3 > HF > CH_4$
(3) $NH_3 > HF > CH_4 > H_2S$
(4) $HF > NH_3 > H_2S > CH_4$

59. Molar conductance of an electrolyte increase with dilution according to the equation:

$$\Lambda_m = \Lambda_m^{o} - A\sqrt{c}$$

Which of the following statements are **true**?

(A) This equation applies to both strong and weak electrolytes.
(B) Value of the constant A depends upon the nature of the solvent.
(C) Value of constant A is same for both $BaCl_2$ and $MgSO_4$
(D) Value of constant A is same for both $BaCl_2$ and $Mg(OH)_2$

Choose the **most appropriate** answer from the options given below:

(1) (A) and (B) only
(2) (A), (B) and (C) only
(3) (B) and (C) only
(4) (B) and (D) only

60. Cheilosis occurs due to deficiency of________.

(1) thiamine
(2) nicotinamide
(3) pyridoxamine
(4) riboflavin

61. The **correct** value of cell potential in volt for the reaction that occurs when the following two half cells are connected, is

$$Fe^{2+}_{(aq)} + 2e^{-} \rightarrow Fe(s),\ E^{\circ} = -0.44\ V$$

$$Cr_2O_7^{2-}{}_{(aq)} + 14H^{+} + 6e^{-} \rightarrow 2Cr^{3+} + 7H_2O,\quad E^{\circ} = +1.33\ V$$

(1) +1.77 V
(2) +2.65 V
(3) +0.01 V
(4) +0.89 V

62. $R–COOH \xrightarrow[\text{(ii)}\,H_2O/HCl]{\text{(i) "X"}} R–CH_2OH$

$R–CH{=}CH_2 \xrightarrow[\text{(ii)}\,H_2O,NaOH,H_2O_2]{\text{(i) "X"}} R–CH_2–CH_2–OH$

Identify 'X' in above reactions

(1) B_2H_6
(2) $LiAlH_4$
(3) $NaBH_4$
(4) H_2/Pd

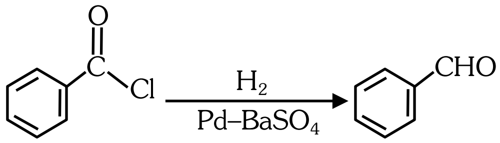
63. For a reaction $3A \rightarrow 2B$

The average rate of appearance of B is given by $\frac{\Delta[B]}{\Delta t}$. The **correct** relation between the average rate of appearance of B with the average rate of disappearance of A is given in option :

(1) $\frac{-\Delta[A]}{\Delta t}$ (2) $\frac{-3\Delta[A]}{2\Delta t}$

(3) $\frac{-2\Delta[A]}{3\Delta t}$ (4) $\frac{\Delta[A]}{\Delta t}$

64. The following conversion is known as :



(1) Stephen reaction
(2) Gattermann-Koch reaction
(3) Etard reaction
(4) Rosenmund reaction

65. Which amongst the following is used in controlling depression and hypertension?

(1) Seldane (2) Valium
(3) Equanil (4) Prontosil

66. Which one of the following represents all isoelectronic species ?

(1) Na^+, Cl^-, O^-, NO^+
(2) N_2O, N_2O_4, NO^+, NO
(3) Na^+, Mg^{2+}, O^-, F^-
(4) Ca^{2+}, Ar, K^+, Cl^-

67. Given below are two statements :

Statement I : The value of wave function, Ψ depends upon the coordinates of the electron in the atom.

Statement II :
The probability of finding an electron at a point within an atom is proportional to the orbital wave function.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement I** is true but **Statement II** is false.
(2) **Statement I** is false but **Statement II** is true.
(3) Both **Statement I** and **Statement II** are true.
(4) Both **Statement I** and **Statement II** are false.

68. The **correct** van der Waals equation for 1 mole of a real gas is :

(1) $\left(p+\frac{a}{V^2}\right)(V-b)=RT$

(2) $\left(p+\frac{V^2}{a}\right)(V-b)=RT$

(3) $\left(p+\frac{an^2}{V^2}\right)(V^2-nb)=RT$

(4) $\left(p+\frac{an^2}{V}\right)(V-nb)=nRT$

69. The **correct** option in which the density of argon (Atomic mass = 40) is highest :

(1) STP (2) 0°C, 2 atm
(3) 0°C, 4 atm (4) 273°C, 4 atm

70. Which of the following is **correctly** matched ?

(1) Basic oxides $\Rightarrow In_2O_3, K_2O, SnO_2$
(2) Neutral oxides $\Rightarrow CO, NO_2, N_2O$
(3) Acidic oxides $\Rightarrow Mn_2O_7, SO_2, TeO_3$
(4) Amphoteric oxides $\Rightarrow BeO, Ga_2O_3, GeO$

71. Which of the following is a positively charged sol?

(1) Methylene blue sol (2) Congo red sol
(3) Silver sol (4) Sb_2S_3 sol

72. Match **List-I** with **List-II**

List-I (Mixtures/Sample)		**List-II (Technique used for purification)**	
(A)	Glycerol from spent lye	(I)	Steam distillation
(B)	Chloroform + Aniline	(II)	Fractional distillation
(C)	Fractions of crude oil	(III)	Distillation under reduced pressure
(D)	Aniline + Water	(IV)	Distillation

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(IV),(C)-(II),(D)-(I)
(2) (A)-(IV), (B)-(II),(C)-(I),(D)-(III)
(3) (A)-(I), (B)-(II),(C)-(III),(D)-(IV)
(4) (A)-(I), (B)-(III),(C)-(II),(D)-(IV)

73. Which amongst the following reactions of alkyl halides produces isonitrile as a major product ?

(A) $R - X + HCN \rightarrow$

(B) $R - X + AgCN \rightarrow$

(C) $R - X + KCN \rightarrow$

(D) $R - X + NaCN \xrightarrow[C_2H_5OH]{H_2O}$

Choose the **most appropriate** answer from the options given below :

(1) (D) only

(2) (C) and (D) only

(3) (B) only

(4) (A) and (B) only

74. The **List-I** with **List-II**

List-I (Hydride)		List-II (Type of Hydride)	
(A)	NaH	(I)	Electron precise
(B)	PH_3	(II)	Saline
(C)	GeH_4	(III)	Metallic
(D)	$LaH_{2.87}$	(IV)	Electron rich

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(2) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(3) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

75. Which one of the following statements is **incorrect** related to Molecular Orbital Theory ?

(1) The π^* antibonding molecular orbital has a node between the nuclei.

(2) In the formation of bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other.

(3) Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are symmetrical around the bond axis.

(4) A π-bonding molecular orbital has larger electron density above and below the internuclear axis.

76. An acidic buffer is prepared by mixing :

(1) weak acid and it's salt with strong base

(2) equal volumes of equimolar solutions of weak acid and weak base

(3) strong acid and it's salt with strong base

(4) strong acid and it's salt with weak base

(The pK_a of acid = pK_b of the base)

77. Reagents which can be used to convert alcohols to carboxylic acids, are

(A) $CrO_3 - H_2SO_4$

(B) $K_2Cr_2O_7 + H_2SO_4$

(C) $KMnO_4 + KOH/H_3O^+$

(D) Cu, 573 K

(E) CrO_3, $(CH_3CO)_2O$

Choose the **most appropriate** answer from the options given below :

(1) (B), (C) and (D) only (2) (B), (D) and (E) only

(3) (A), (B) and (C) only (4) (A), (B) and (E) only

78. Select the element (M) whose trihalides **cannot** be hydrolysed to produce an ion of the form $[M(H_2O)_6]^{3+}$

(1) Ga (2) In (3) Al (4) B

79. The **correct** options for the rate law that corresponds to overall first order reaction is

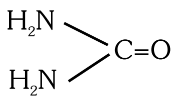
(1) Rate = $k[A]^0 [B]^2$

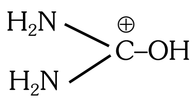
(2) Rate = $k[A] [B]$

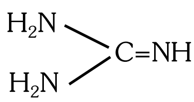
(3) Rate = $k[A]^{1/2} [B]^2$

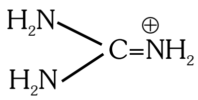
(4) Rate = $k[A]^{-1/2} [B]^{3/2}$

80. Which amongst the following compounds/species is least basic ?

(1) 

(2) 

(3) 

(4) 

81. Which of the following forms a set of complex and a double salt, respectively ?

(1) $CuSO_4.5H_2O$ and $CuCl_2,4NH_3$

(2) $PtCl_2.2NH_3$ and $PtCl_4.2HCl$

(3) $K_2PtCl_2.2NH_3$ and $KAl(SO_4)_2.12H_2O$

(4) $NiCl_2.6H_2O$ and $NiCl_2(H_2O)_4$

82. Given below are two statements:

Statement I :

High density polythene is formed in the presence of catalyst triethylaluminium and titanium tetrachloride.

Statement II :

High density polymers are chemically inert.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement-I** is correct but **Statement-II** is false.

(2) **Statement-I i**s incorrect but **Statement-II** is true.

(3) Both **Statement-I** and **Statement-II** are true.

(4) Both **Statement-I** and **Statement-II** are false.

83. Which amongst the following compounds will show geometrical isomerism ?

(1) Pent-1-ene

(2) 2,3-Dimethylbut-2-ene

(3) 2-Methylprop-1-ene

(4) 3,4-Dimethylhex-3-ene

84. Given below are two statements:

Statement I :

Hydrated chlorides and bromides of Ca, Sr and Ba on heating undergo hydrolysis.

Statement II :

Hydrated chlorides and bromides of Be and Mg on heating undergo dehydration.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement-I** is correct but **Statement-II** is false.

(2) **Statement-I** is incorrect but **Statement-II** is true.

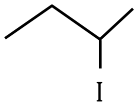
(3) Both **Statement-I** and **Statement-II** are true.

(4) Both **Statement-I** and **Statement-II** are false.

85. The correct order for the rate of α, β-dehydrohalogenation for the following compounds is ____.

(i) 

(ii) 

(iii) 

(1) (i) < (ii) < (iii) (2) (ii) < (i) < (iii)

(3) (iii) < (ii) < (i) (4) (ii) < (iii) < (i)

Section-B (Chemistry)

86. How many number of tetrahedral voids are formed in 5 mol of a compound having cubic close packed structure ? (Choose the **correct** option)

(1) 1.550×10^{24} (2) 3.011×10^{25}

(3) 3.011×10^{24} (4) 6.022×10^{24}

87. Type of isomerism exhibited by compounds $[Cr(H_2O)_6Cl_3$, $[Cr(H_2O)_5Cl]Cl_2.H_2O$, $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ and the value of coordination number (CN) of central metal ion in all these compounds, respectively is :

(1) Geometrical isomerism, CN = 2

(2) Optical isomerism, CN = 4

(3) Ionisation isomerism, CN = 4

(4) Solvate isomerism, CN = 6

88. The **correct** sequence given below containing neutral, acidic, basic and amphoteric oxide each, respectively, is

(1) NO, ZnO, CO_2, CaO

(2) ZnO, NO, CaO, CO_2

(3) NO, CO_2, ZnO, CaO

(4) NO, CO_2, CaO, ZnO

89. Read the following statements and choose the set of **correct** statements :

(A) Chrome steel is used for cutting tools and crushing machines.

(B) The fine dust of aluminium is used in paints and lacquers.

(C) Copper is used for reduction of alcohol

(D) Zinc dust is used as a reducing agent in the manufacture of paints

(E) Iron is used for galvanising zinc

Choose the **most appropriate** answer from the options given below :

(1) (D) and (E) only (2) (A) and (D) only

(3) (A), (B) and (D) only (4) (B), (C) and (D) only

90. Choose the **correct** sequence of reagents in the conversion of 4-nitrotoluene to 2-bromotoluene.

(1) $NaNO_2/HCl$; Sn/HCl; Br_2; H_2O/H_3PO_2

(2) Sn/HCl ; $NaNO_2/HCl$; Br_2;H_2O/H_3PO_2

(3) Br_2;Sn/HCl;$NaNO_2/HCl$;H_2O/H_3PO_2

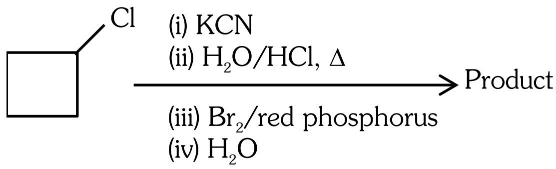
(4) Sn/HCl;Br_2;$NaNO_2/HCl$;H_2O/H_3PO_2

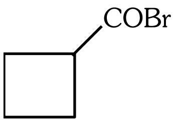
91. How are edge length 'a' of the unit cell and radius 'r' of the sphere related to each other in ccp structure? (Choose **correct** option for your answer)

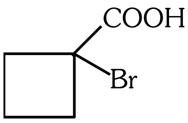
(1) $a = 2r$ (2) $a = r/\ 2\sqrt{2}$

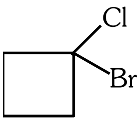
(3) $a = 4r/\ \sqrt{3}$ (4) $a = 2\sqrt{2}\ r$

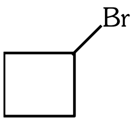
92. Identify the product in the following reaction



(1) 

(2) 

(3) 

(4) 

93. Given below are **two** statements:

Statement I :

In an organic compound, when inductive and electromeric effects operate in opposite directions, the inductive effect predominates.

Statement II :

Hyperconjugation is observed in o-xylene.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement-I** is true but **Statement-II** is false.

(2) **Statement-I** is false but **Statement-II** is true.

(3) Both **Statement-I** and **Statement-II** are true.

(4) Both **Statement-I** and **Statement-II** are false.

94. The **correct** option for a redox couple is :

(1) Both are oxidised forms involving same element.

(2) Both are reduced forms involving same element.

(3) Both the reduced and oxidized forms involve same element.

(4) Cathode and anode together.

95. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) :- Ionisation enthalpies of early actinoids are lower than for early lanthanoids.

Reason (R) : Electrons are entering 5f orbitals in actinoids which experience greater shielding from nuclear charge.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **(A)** is true but **(R)** is false.

(2) **(A)** is false but **(R)** is true

(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(4) Both **(A)** and **(R)** are true but (R) is **not** the correct explanation of **(A)**.

96. Consider the following reaction :-

$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)\ \Delta_rH° = -483.64$ kJ.

What is the enthalpy change for decomposition of one mole of water ? (Choose the **right** option).

(1) 120.9 kJ (2) 241.82 kJ

(3) 18 kJ (4) 100 kJ

97. Which statement is **not** true about photochemical smog?

(1) Photochemical smog is harmful to humans but has no effect on plants.

(2) Plants like Pinus, Juniparus can help in reducing the photochemical smog.

(3) Photochemical smog occurs in warm, dry and sunny climate.

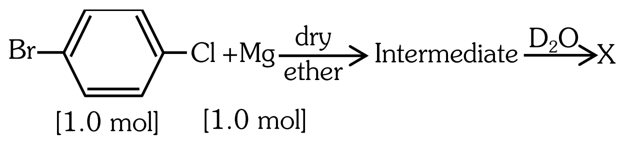
(4) Common components of photochemical smog are ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate.

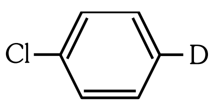
98. Which amongst the following aqueous solution of electrolytes will have minimum elevation in boiling point? Choose the **correct** option :-

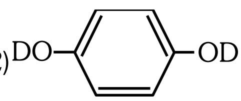
(1) 0.05 M NaCl (2) 0.1 M KCl

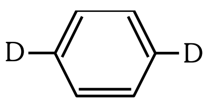
(3) 0.1 M $MgSO_4$ (4) 1 M NaCl

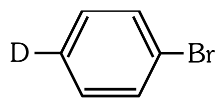
99. Identify 'X' in the following reaction.



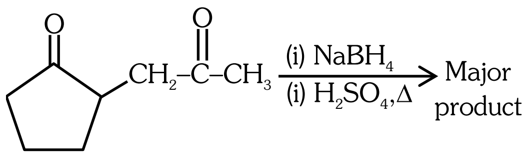
(1) 

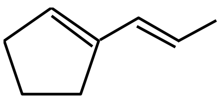
(2) 

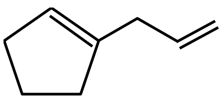
(3) 

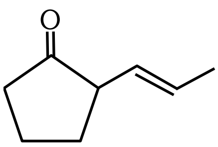
(4) 

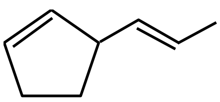
100. The **major** product formed in the following conversion is _____.



(1) 

(2) 

(3) 

(4) 

Section - A (Biology : Botany)

101. Match List-I with List-II

List-I	**List-II**
(A) Protein	(I) C=C double bonds
(B) Unsaturated fatty acid	(II)Phosphodiester bond
(C) Nucleic acid	(III) Glycosidic bonds
(D) Polysaccharide	(IV) Peptide bonds

Choose the **correct** answer from the options given below :

(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(4) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

102. Match List-I with List-II.

	List-I		**List-II**
(A)	Hydrarch succession	(I)	Gradual change in the species composition
(B)	Xerarch succession	(II)	Faster and climax reached quickly
(C)	Ecological succession	(III)	Lichens to mesic conditions
(D)	Secondary succession	(IV)	Phytoplankton to mesic conditions

Choose the **correct** answer from the options given below :

(1) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

103. In *Calotropis*, aestivation is :

(1) Valvate (2) Vexillary

(3) Imbricate (4) Twisted

104. Match List-I with List-II.

	List-I		**List-II**
(A)	Chlorophyll a	(I)	Yellow to yellow orange
(B)	Chlorophyll b	(II)	Yellow green
(C)	Xanthophyll	(III)	Blue green
(D)	Carotenoid	(IV)	Yellow

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

105. Nitrates and phosphates flowing from agricultural farms into water bodies are a significant cause of:

(1) Eutrophication

(2) Humification

(3) Mineralisation

(4) Stratification

106. Match List-I with List-II.

	List-I (Type of cross)		List-II (Phenotypic ratio)
(A)	Monohybrid Cross	(I)	1 : 1
(B)	Dihybrid Cross	(II)	1 : 2 : 1
(C)	Incomplete dominance	(III)	3 : 1
(D)	Test Cross	(IV)	9 : 3 : 3 : 1

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(2) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)
(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

107. How many times decarboxylation occurs during each TCA cycle ?

(1) Thrice (2) Many
(3) Once (4) Twice

108. The dissolution of synaptonemal complex occurs during :

(1) Pachytene (2) Diplotene
(3) Diakinesis (4) Leptotene

109. Identify the **correct** statements regarding Mass flow hypothesis.

(A) Mass flow is faster than diffusion.
(B) Mass flow is the result of pressure difference between the end points.
(C) Different substances involved in mass flow move at different paces.
(D) Mass flow can result through either a positive or a negative hydrostatic pressure gradient.

Choose the **correct** answer from the options given below :

(1) (A), (C), (D) only
(2) (B), (C), (D) only
(3) (A), (B), (C) only
(4) (A), (B), (D) only

110. Doubling of the number of chromosomes can be achieved by disrupting mitotic cell division soon after :

(1) Anaphase
(2) Telophase
(3) Prophase
(4) Metaphase

111. Given below are two statements :

Statement I :
RuBisCO is the most abundant enzyme in the world.

Statement II :
Photorespiration does not occur in C_4 plants.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect
(2) **Statement I** is incorrect but **Statement II** is correct
(3) Both **Statement I** and **Statement II** are correct
(4) Both **Statement I** and **Statement II** are incorrect

112. In 'rivet popper hypothesis', Paul Ehrlich compared the rivets in an airplane to :

(1) species within a genus
(2) genetic diversity
(3) ecosystem
(4) genera within a family

113. In a pea flower, five petals are arranged in a specialized manner with one posterior, two lateral and two anterior. These are named as _______, _______ and ________ respectively.

(1) Keel, Wings and Standard
(2) Vexillum, Keel and Standard
(3) Keel, Standard and Carina
(4) Standard, Wings and Keel

114. In which of the following sets of families, the pollen grains are viable for months ?
(1) Solanaceae, Poaceae and Liliaceae
(2) Brassicaceae, Liliaceae and Poaceae
(3) Rosaceae, Liliaceae and Poaceae
(4) Leguminosae, Solanaceae and Rosaceae

115. Transfer of pollen grains from anther to stigma of another flower of same plant is known as :
(1) Geitonogamy (2) Xenogamy
(3) Autogamy (4) Cleistogamy

116. The phenomenon which is influenced by auxin and also played a major role in its discovery :
(1) Phototropism (2) Root initiation
(3) Gravitropism (4) Apical Dominance

117. The transverse section of a plant part showed polyarch, radial and exarch xylem, with endodermis and pericycle. The plant part is identified as :
(1) Monocot root (2) Dicot root
(3) Dicot stem (4) Monocot stem

118. What will happen if fresh water lake gets contaminated by addition of polluted water with high BOD ?
(1) Amount of dissolved oxygen in the lake will decrease
(2) The lake will remain unaffected
(3) Number of submerged aquatic plants in the lake will increase
(4) Number of aquatic animals in the lake will increase

119. The last chromosome sequenced in Human Genome Project was :
(1) Chromosome 6 (2) Chromosome 1
(3) Chromosome 22 (4) Chromosome 14

120. The amount of nutrients such as carbon, nitrogen, potassium and calcium present in the soil at any given time is referred to as :
(1) Standing state (2) Standing crop
(3) Humus (4) Detritus

121. Plants offer rewards to animals in the form of pollen and nectar and the animals facilitate the pollination process. This is an example of :
(1) Amensalism (2) Competition
(3) Commensalism (4) Mutualism

122. The species of plants that plays a vital role in controlling the relative abundance of other species in a community is called _____ .
(1) alien species (2) endemic species
(3) exotic species (4) keystone species

123. Match List-I List-II.

	List-I		List-II
(A)	Pteropsida	(I)	*Psilotum*
(B)	Lycopsida	(II)	*Equisetum*
(C)	Psilopsida	(III)	*Adiantum*
(D)	Sphenopsida	(IV)	*Selaginella*

Choose the **correct** answer from the options given below :
(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

124. Inulin is a polymer of :
(1) Fructose (2) Galactose
(3) Amino acids (4) Glucose

125. Thermostable DNA polymerase used in PCR was isolated from :
(1) *Thermus aquaticus*
(2) *Escherichia coli*
(3) *Agrobacterium tumifaciens*
(4) *Bacillus thuringiensis*

126. Name the component that binds to the operator region of an operon and prevents RNA polymerase from transcribing the operon.
(1) Promotor
(2) Regulator protein
(3) Repressor protein
(4) Inducer

127. A heterozygous pea plant with violet flowers was crossed with homozygous pea plant with white flower. Violet is dominant over white. Which one of the following represents the expected combinations among 40 progenies formed ?
(1) 30 produced violet and 10 produced white flowers
(2) 20 produced violet and 20 produced white flowers
(3) All 40 produced violet flowers
(4) All 40 produced white flowers

128. Fatty acids are connected with the respiratory pathway through :
(1) Acetyl CoA
(2) α-Ketoglutaric acid
(3) Dihydroxy acetone phosphate
(4) Pyruvic acid

129. Ligation of foreign DNA at which of the following site will result in loss of tetracyclin resistance of pBR322 :
(1) Pst I (2) Pvu I
(3) EcoR I (4) BamH I

130. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Auxin	(I)	Promotes female flower formation in cucumber
(B)	Gibberellin	(II)	Overcoming apical dominance
(C)	Cytokinin	(III)	Increase in the length of grape stalks
(D)	Ethylene	(IV)	Promotes flowering in pineapple

Choose the **correct** answer from the options given below :
(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(3) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

131. During symport two different molecules move across the membrane :
(1) in same direction with the help of different carriers located at a common site
(2) in same direction with the help of different carriers located at different sites in the same cell
(3) in same direction with the help of same carrier
(4) in opposite direction with the help of same carrier

132. Which classes of algae possess pigment fucoxanthin and pigment phycoerythrin, respectively ?
(1) Phaeophyceae and Chlorophyceae
(2) Phaeophyceae and Rhodophyceae
(3) Chlorophyceae and Rhodophyceae
(4) Rhodophyceae and Phaeophyceae

133. In which disorder change of single base pair in the gene for beta globin chain results in change of glutamic acid to valine ?
(1) Thalassemia
(2) Sickle cell anemia
(3) Haemophilia
(4) Phenylketonuria

134. For chemical defence against herbivores, *Calotropis* has :
(1) cardiac glycosides (2) strychnine
(3) toxic ricin (4) distasteful quinine

135. Consider the following tissues in the stelar region of a stem showing secondary growth.
(A) Primary xylem
(B) Secondary xylem
(C) Primary phloem
(D) Secondary phloem
Arrange these in the **correct** sequence of their position from pith towards corts.
(1) (A), (B), (D), (C) (2) (B), (A), (C), (D)
(3) (A), (B), (C), (D) (4) (B), (A), (D), (C)

Section - B (Biology : Botany)

136. Which of the following mineral ion is not remobilized in plants ?

(1) Potassium (2) Calcium

(3) Nitrogen (4) Phosphorus

137. Which out of the following statements is incorrect ?

(1) Grana lamellae have both PS I and PS II

(2) Cyclic photophosphorylation involves both PS I and PS II

(3) Both ATP and NADPH + H^+ are synthesised during non-cyclic photophosphorylation.

(4) Stroma lamellae lack PS II and NADP reductase

138. Match **Column-I** with **Column-II**.

	Column-I		Column-II
(A)	*Nitrococcus*	(I)	Denitrification
(B)	*Rhizobium*	(II)	Conversion of ammonia to nitrite
(C)	*Thiobacillus*	(III)	Conversion of nitrite to nitrate
(D)	*Nitrobacter*	(IV)	Conversion of atmospheric nitrogen to ammonia

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(3) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(4) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

139. In angiosperms the correct sequence of events in formation of female gametophyte in the ovule is :

(A) 3 successive free nuclear divisions functional megaspore.

(B) Degeneration of 3 megaspores.

(C) Meiotic division in megaspore mother cell.

(D) Migration of 3 nuclei towards each pole.

(E) Formation of wall resulting in seven celled embryosac.

Choose the **correct** answer from the options given below :

(1) (A), (B), (C), (D), (E) (2) (C), (E), (A), (D), (B)

(3) (B), (C), (A), (D), (E) (4) (C), (B), (A), (D), (E)

140. Which of the following statements is true ?

(1) All pteridophytes exhibit haplo-diplontic pattern.

(2) Seed bearing plants follow haplontic pattern

(3) Most algal genera are diplontic

(4) Most bryophytes do not have haplo-diplontic life cycle.

141. Which of the following statement is **incorrect** about *Agrobacterium tumifaciens* ?

(1) It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.

(2) 'Ti' plasmid from *Agrobacterium tumifaciens* used for gene transfer is not pathogenic to plant cell.

(3) It transforms normal plant cells into tumor cells.

(4) It delivers 'T-DNA' into plant cell.

142. Consider the following plant tissues :

(A) Axillary buds

(B) Fascicular vascular cambium

(C) Interfascicular cambium

(D) Cork cambium

(E) Intercalary meristem

Identify the lateral meristems among the above.

(1) (A), (C) and (D) only

(2) (B), (C) and (D) only

(3) (A), (B), (C) and (E) only

(4) (A), (B), (D) and (E) only

143. Match **List-I** with **List-II.**

	List-I		List-II
(A)	Kanamycin	(I)	Delivers genes into animal cells
(B)	ClaI	(II)	Selectable marker
(C)	Disarmed retroviruses	(III)	Restriction site
(D)	Kanamycin^R gene	(IV)	Antibiotic resistance

Choose the **correct** answer from the options given below :

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

144. Given below are two statements :

Statement I :

The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

Statement II :

A transcription unit in DNA is defined primarily by the three regions in the DNA i.e., a promotor, the structural gene and a terminator.

In the light of the above statements, choose the correct answer from the options given below :

(1) **Statement I** is true but **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true

(4) Both **Statement I** and **Statement II** are false

145. Which scientist conducted an experiment with ^{32}P and ^{35}S labelled phages for demonstrating that DNA is the genetic material ?

(1) james D. Watson and F.H. C. Crick

(2) A. D Hershey and M.J. Chase

(3) F. Griffith

(4) O.T. Avery, C.M. MacLeod and M. McCarty

146. A certain plant homozygous for yellow seeds and red flowers was crossed with a plant homozygous for green seeds and white flowers. The F_1 plants had yellow seeds and pink flowers. The F_1 plants were selfed to get F_2 progeny. Assuming independent assortment of the two characters, how many phenotypic categories are expected for these characters in the F_2 generation ?

(1) 9 (2) 16 (3) 4 (4) 6

147. During which stages of mitosis and meiosis, respectively does the centromere of each chromosome split ?

(1) Mataphase, Metaphase II

(2) Prophase, Telophase I

(3) Telophase, Anaphase I

(4) Anaphase, Anaphase II

148. Which os the following statements is **not correct** ?

(1) Phase of cell elongation of plant cells is characterized by increased vacuolation.

(2) Cells in the meristematic phase of growth exhibit abundant plasmodesmatal connections.

(3) Plant growth is generally determinate.

(4) Plant growth is measurable.

149. Match the following :

	Type of flower		Example
(A)	Zygomorphic	(I)	Mustard
(B)	Hypogynous	(II)	Plum
(C)	Perigynous	(III)	*Cassia*
(D)	Epigynous	(IV)	Cucumber

Select the **correct** option :

(1) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(2) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(3) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

150. Given below are two statements :

Statement I :

The process of translocation through phloem is unidirectional but through xylem, it is bidirectional.

Statement II :

The most readily mobilized elements are phosphorus, sulphur, nitrogen and potassium.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

Section - A (Biology : Zoology)

151. Which of the following sexually transmitted infections are completely curable ?

(1) HIV infection and Trichomoniasis
(2) Syphilis and trichomoniasis
(3) Hepatitis – B and Genital herpes
(4) Genital herpes and Genital warts

152. Match **List - I** with **List - II**.

	List-I		**List-II**
(A)	Typhoid	(I)	Protozoan
(B)	Elephantiasis	(II)	*Salmonella*
(C)	Ringworm	(III)	Aschelminthes
(D)	Malaria	(IV)	*Microsporum*

Choose the **correct** answer from the options given below :

(1) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)
(2) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
(4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

153. Which of the following is not a secondary metabolite?

(1) Curcumin (2) Morphine
(3) Anthocyanin (4) Lecithin

154. Arrange the sequence of different hormones for their role during gametogenesis.

(A) Gonadotropin LH stimulates synthesis and secretion of Androgen
(B) Gonadotropin releasing hormone from hypothalamus
(C) Androgen stimulates spermatogenesis
(D) Gonadotropin FSH helps in the process of spermiogenesis
(E) Gonadotropins from anterior pituitary gland.

Choose the **correct** answer from the options given below :

(1) (E), (A), (D), (B), (C) (2) (C), (A), (D), (E), (B)
(3) (B), (E), (A), (C), (D) (4) (D), (B), (A), (C), (E)

155. House fly belongs to _____ family.

(1) Cyprinidae (2) Hominidae
(3) Calliphoridae (4) Muscidae

156. Select **incorrect** statement, regarding chemical structure of insulin.

(1) Mature insulin molecule consists of three polypeptide chains-A, B and C.
(2) Insulin is synthesized as prohormone which contains extra stretch of C-peptide.
(3) C-peptide is not present in mature insulin molecule.
(4) Polypeptide chains A and B are linked by disulphide bridges.

157. Which one of the following is the quiescent stage of cell cycle ?

(1) M (2) G_2 (3) G_1 (4) G_0

158. Given below are two statements :

Statement I :
RNA being unstable, mutate at a faster rate.

Statement II :
RNA can directly code for synthesis of proteins hence can easily express the characters.

In the light of the above statements, choose the correct answer from the options given below :

(1) **Statement I** is correct but **Statement II** is false
(2) **Statement I** is incorrect but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false

159. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A) :
Ascending limb of loop of Henle is impermeable to water and allows transport of electrolytes actively or passively.

Reason (R) :
Dilution of filtrate takes place due to efflux of electrolytes in the medullary fluid.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **(A)** is true but **(R)** is false
(2) **(A)** is false but **(R)** is true
(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
(4) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**

160. The Cockroach is :

(1) Ammonotelic only

(2) Uricotelic only

(3) Ureotelic only

(4) Ureotelic and Uricotelic

161. Which of the following statements are **correct** with respect to the hormone and its function ?

(A) Thyrocalcitonin (TCT) regulates the blood calcium level.

(B) In males, FSH and androgens regulate spermatogenesis.

(C) Hyperthyroidism can lead to goitre.

(D) Glucocorticoids are secreted in Adrenal Medulla.

(E) Parathyroid hormone is regulated by circulating levels of sodium ions.

Choose the **most appropriate** answer from the options given below :

(1) (C) and (E) only (2) (A) and (B) only

(3) (B) and (C) only (4) (A) and (D) only

162. Select the sequence of steps in Respiration.

(A) Diffusion of gases (O_2 and CO_2) across alveolar membrane.

(B) Diffusion of O_2 and CO_2 between blood and tissues.

(C) Transport of gases by the blood

(D) Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.

(E) Utilisation of O_2 by the cells for catabolic reactions are resultant release of CO_2

Choose the **correct** answer from the options given below :

(1) (D), (A), (C), (B), (E)

(2) (C), (B), (A), (E), (D)

(3) (B), (C), (E), (D), (A)

(4) (A), (C), (B), (E), (D)

163. Which of the following is/are cause(s) of biodiversity losses ?

(1) Over-exploitation, habitat loss and fragmentation.

(2) Climate change only

(3) Over-Exploitation only

(4) Habitat loss and fragmentation only

164. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Contractile vacuole	(I)	*Asterias*
(B)	Water vascular system	(II)	*Amoeba*
(C)	Canal system	(III)	*Spongilla*
(D)	Flame cells	(IV)	*Taenia*

Choose the **correct** answer from the options given below :

(1) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(2) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(3) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(4) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

165. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Palm bones	(I)	Phalanges
(B)	Wrist bones	(II)	Metacarpals
(C)	Ankle bones	(III)	Carpals
(D)	Digit bones	(IV)	Tarsals

Choose the **correct** answer from the options given below :

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(3) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

166. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Non-medicated IUDs	(I)	Multiload 375
(B)	Copper releasing IUDs	(II)	Rubber barrier
(C)	Hormone releasing IUDs	(III)	Lippes loop
(D)	Vaults	(IV)	LNG-20

Choose the correct answer from the options given below :

(1) A-(IV), B-(III), C-(I), D-(II)

(2) A-(II), B-(IV), C-(III), D-(I)

(3) A-(III), B-(I), C-(IV), D-(II)

(4) A-(III), B-(IV), C-(II), D-(I)

167. Which of the following can act as molecular scissors?
(1) Restriction enzymes
(2) DNA ligase
(3) RNA polymerase
(4) DNA polymerase

168. Select the **correct** statements about sickle cell anaemia.
(A) There is a change in gene for beta globin.
(B) In the beta globin, there is valine in the place of Lysine.
(C) It is an example of point mutation.
(D) In the normal gene U is replaced by A.
Choose the **correct** answer from the options given below :
(1) (B), (C) and (D) only
(2) (B) and (D) only
(3) (A), (B) and (D) only
(4) (A) and (C) only

169. Given below are two statements :
Statement I :-
Intra Cytoplasmic Sperm Injection (ICSI) is another specialised procedure of in-vivo fertilisation.
Statement II :-
Infertility cases due to inability of the male partner to inseminate female can be corrected by artificial insemination (AI).
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **Statement I** is correct but **statement II** is false
(2) **Statement I** is incorrect but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false.

170. Match **List-I** with **List-II**.

	List-I (ECG)		**List-II (Electrical activity of heart)**
(A)	P-wave	(I)	Depolarisation of ventricles
(B)	QRS complex	(II)	End of systole
(C)	T wave	(III)	Depolarisation of atria
(D)	End of T wave	(IV)	Repolarisation of ventricles

Choose the **correct** answer from the options given below :
(1) A-(IV), B-(I), C-(III), D-(II)
(2) A-(I), B-(IV), C-(III), D-(II)
(3) A-(IV), B-(III), C-(I), D-(II)
(4) A-(III), B-(I), C-(IV), D-(II)

171. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Eosinophils	(I)	6 – 8%
(B)	Lymphocytes	(II)	2 – 3%
(C)	Neutrophils	(III)	20 – 25%
(D)	Monocytes	(IV)	60 – 65 %

Choose the **correct** answer from the options given below :
(1) A-(IV), B-(I), C-(II), D-(III)
(2) A-(IV), B-(I), C-(III), D-(II)
(3) A-(II), B-(III), C-(IV), D-(I)
(4) A-(II), B-(III), C-(I), D-(IV)

172. Given below are two statements :
Statement I :-
Goblet cells are unicellular glands.
Statement II :-
Earwax is the secretion of exocrine gland.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **Statement I** is true but **Statement II** is false
(2) **Statement I** is false but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false.

173. Given below are two statements regarding oogenesis:

Statement I :-

The primary follicles get surrounded by more layers of granulosa cells, a theca and shows fluid filled cavity antrum. Now it is called secondary follicle.

Statement II :-

Graffian follicle ruptures to release the secondary oocyte from the ovary by the process called ovulation.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **Statement I** is correct but **Statement II** is false.
(2) **Statement I** is incorrect but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false.

174. If there are 250 snails in a pond, and within a year their number increases to 2500 by reproduction. What should be their birth rate per snail per year ?

(1) 10 (2) 9 (3) 25 (4) 15

175. Given below are two statements :

Statement I :-

The nose contains mucus – coated receptors which are specialised for receiving the sense of smell and are called olfactory receptors.

Statement II :-

Wall of the eye ball has three layers. The external layer is called choroid (dense connective tissue), middle layer is sclera (thin pigmented layer) and internal layer is retina (ganglion cells, bipolar cells and photoreceptor cells).

In the light of the above statements, choose the correct answer from the options given below:

(1) **Statement I** is true but **statement II** is false
(2) **Statement I** is false but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false.

176. Which one of the following acts as an inducer for lac operon ?

(1) Sucrose (2) Lactose
(3) Glucose (4) Galactose

177. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Deforestation	(I)	Responsible for heating of Earth's surface and atmosphere
(B)	Reforestation	(II)	Conversion of forested areas to non-forested areas
(C)	Green-house effect	(III)	Natural ageing of lake by nutrient enrichment of its water
(D)	Eutrophication	(IV)	Process of restoring a forest that once existed but was removed

Choose the **correct** answer from the options given below :

(1) A-(IV), B-(III), C-(II), D-(I)
(2) A-(I), B-(II), C-(III), D-(IV)
(3) A-(III), B-(I), C-(II), D-(IV)
(4) A-(II), B-(IV), C-(I), D-(III)

178. Diacetyl morphine is also called as :

(1) Amphetamine (2) Barbiturate
(3) Crack (4) Smack

179. 'X' and 'Y' are the components of Binomial nomenclature. This naming system was proposed by 'Z' :

(1) X-Generic name, Y-Specific epithet, Z-Carolus Linnaeus
(2) X-Specific epithet, Y-Generic name, Z-R.H. Whittaker
(3) X-Specific epithet, Y-Generic name, Z-Carolus Linnaeus
(4) X-Generic name, Y-Specific epithet, Z-R.H. Whittaker

180. Which of the following statements are **correct** ?

(A) Reproductive health refers to total well-being in all aspects of reproduction.

(B) Amniocentesis is legally banned for sex determination in India.

(C) "Saheli" – a new oral contraceptive for females was developed in collaboration with ICMR (New Delhi).

(D) Amniocentesis is used to determine genetic disorders and survivability of foetus.

Choose the **most appropriate** answer from the options given below :

(1) (B) and (C) only (2) (D) and (C) only

(3) (A), (B) and (D) only (4) (A) and (C) only

181. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Terpenoides	(I)	Codeine
(B)	Lectins	(II)	Diterpenes
(C)	Alkaloids	(III)	Ricin
(D)	Toxins	(IV)	Concanavalin A

Choose the **correct** answer from the options given below :

(1) A-(II), B-(IV), C-(III), D-(I)

(2) A-(II), B-(I), C-(IV), D-(III)

(3) A-(II), B-(III), C-(I), D-(IV)

(4) A-(II), B-(IV), C-(I), D-(III)

182. Given below are two statements:

Statement I :-

In bacteria, the mesosomes are formed by the extensions of plasma membrane.

Statement II :-

The mesosomes, in bacteria, help in DNA replication and cell wall formation.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

183. Select **correct** sequence of substages of Prophase-I of Meiotic division :

(A) Zygotene (B) Pachytene

(C) Diakinesis (D) Leptotene

(E) Diplotene

Choose the **correct** answer from the options given below :

(1) (D), (B), (A), (E), (C) (2) (A), (B), (D), (E), (C)

(3) (D), (A), (B), (E), (C) (4) (A), (D), (B), (C), (E)

184. Brainstem of human brain consists of :

(1) Mid-brain, Pons and Medulla Oblongata

(2) Forebrain, Cerebellum and Pons

(3) Thalamus, Hypothalamus and Corpora quadrigemina

(4) Amygdala, Hippocampus and Corpus Callosum

185. Identify the fossil of man who showed the following characteristics.

(A) Brain capacity of 1400 cc

(B) Used hides to protect their body

(C) Buried their dead bodies

In the light of above statements, choose the **correct** answer from the options given below :

(1) *Homo erectus* (2) Neanderthal man

(3) *Homo habilits* (4) *Australopithecus*

Section - B (Biology : Zoology)

186. With reference to Hershey and Chase experiments. Select the **correct** statements.

(A) Viruses grown in the presence of radioactive phosphorus contained radioactive DNA.

(B) Viruses grown on radioactive sulphur contained radioactive proteins.

(C) Viruses grown on radioactive phosphorus contained radioactive protein.

(D) Viruses grown on radioactive sulphur contained radioactive DNA.

(E) Viruses grown on radioactive protein contained radioactive DNA.

Choose the **most appropriate** answer from the options given below :

(1) (D) and (E) only (2) (A) and (B) only

(3) (A) and (C) only (4) (B) and (D) only

187. Select the **correct** sequential steps regarding absorption of fatty acids and glycerol, in intestine.
(A) Micelles are reformed into small protein coated fat globules called chylomicrons.
(B) Micelles move into intestinal mucosa.
(C) Fatty acids and glycerol are incorporated into small droplets called micelles.
(D) Lacteals release the absorbed substances into blood stream.
(E) Chylomicrons are transported into lacteals.
Choose the **correct** answer from the options given below :
(1) (A), (E), (B), (D), (C) (2) (D), (E), (B), (C), (A)
(3) (C), (B), (A), (E), (D) (4) (B), (C), (E), (A), (D)

188. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) :
A person goes to high altitude and experiences "Altitude Sickness" with symptoms like breathing difficulty and heart palpitations.
Reason (R) :
Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) **(A)** is true but **(R)** is false
(2) **(A)** is false but **(R)** is true
(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(4) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

189. The salient features of genetic code are :
(A) The code is palindromic
(B) UGA act as initiator codon
(C) The code is unambiguous and specific
(D) The code is nearly universal
Choose the **most appropriate** answer from the options given below :
(1) (A) and (D) only (2) (B) and (C) only
(3) (A) and (B) only (4) (C) and (D) only

190. Arrange the events of Renin - Angiotensin mechanism in **correct** sequence.
(A) Activation of JG cells and release of renin.
(B) Angiotensin II activates release of aldosterone.
(C) Fall in glomerular blood pressure.
(D) Reabsorption of Na^+ and water from distal convoluted tubule.
(E) Angiotensinogen is converted to Angiotensin I and then to Angiotensin II.
Choose the **correct** answer from the options given below :
(1) (C), (A), (E), (B), (D) (2) (A), (D), (E), (C), (B)
(3) (A), (D), (C), (B), (E) (4) (B), (A), (E), (D), (C)

191. Select the **correct** statements regarding dissolved Oxygen and Biochemical Oxygen Demand.
(A) BOD is inversely related to dissolved oxygen.
(B) Low dissolved oxygen and high BOD lead to loss of aquatic life.
(C) High BOD leads to high dissolved oxygen.
(D) Both BOD and dissolved oxygen are indicator of health of a water body.
(E) Both BOD and dissolved oxygen are affected by amount of organic matter in the water body.
Choose the **most appropriate** answer from the options given below :
(1) (A), (B), (C), (E) only
(2) (A), (B), (D), (E) only
(3) (A), (B), (C), (D) only
(4) (B), (C), (D), (E) only

192. Given below are two statements :
Statement I :
Parathyroid hormone acts on bones and stimulates the process of bone resorption.
Statement II :
Parathyroid hormone along with Thyrocalcitonin plays a significant role in carbohydrate metabolism.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) **Statement I** is correct but **Statement II** is false
(2) **Statement I** is incorrect but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true.
(4) Both **Statement I** and **Statement II** are false.

193. Select the **correct** statements :
(A) Platyhelminthes are triploblastic pseudocoelomate and bilaterally symmetrical organisms.
(B) Ctenophores reproduce only sexually and fertilization is external.
(C) In tapeworm, fertilization is internal but sexes are not separate.
(D) Ctenophores are exclusively marine, diploblastic and bioluminescent organisms.
(E) In sponges, fertilization is external and development is direct.
Choose the **correct** answer from the options given below :
(1) (A), (C) and (D) only
(2) (B), (C) and (D) only
(3) (A) and (E) only
(4) (B) and (D) only

194. Match **List-I** with **List-II**.

List-I	**List-II**
(A) Gene therapy	(I) Separation of DNA fragments
(B) RNA interference	(II) Diagnostic test for AIDS
(C) ELISA	(III) Cellular defence
(D) Gel Electrophoresis	(IV) Allows correction of a gene defect.

Choose the **correct** answer from the options given below :
(1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(2) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

195. Which of the following statements are **correct** with respect of Golgi apparatus ?
(A) It is the important site of formation of glycoprotein and glycolipids.
(B) It produces cellular energy in the form of ATP.
(C) It modifies the protein synthesized by ribosomes on ER.
(D) It facilitates the transport of ions.
(E) It provides mechanical support.
Choose the **most appropriate** answer from the options given below :
(1) (B) and (C) only (2) (A) and (C) only
(3) (A) and (D) only (4) (D) and (E) only

196. Select the **incorrect** statement with respect to Multiple Ovulation Embryo Transfer (MOET) Technology.
(1) Fertilised eggs at 4 to 6 cells - stages are recovered non-surgically from super-ovulating cow and transferred to surrogate mother.
(2) It is used to increase herd size in a short time
(3) Cow is administered with hormones to induce super-ovulation.
(4) Super-ovulating cow is either mated with elite bull or is artificially inseminated.

197. Given below are two statements :
Statement I :
In cockroach, the forewings are transparent and prothoracic in origin.
Statement II :
In cockroach, the hind wings are opaque, leathery and mesothoracic in origin.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) **Statement I** is correct but **Statement II** is false
(2) **Statement I** is incorrect but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false

198. Match **List-I** with **List-II**.

List-I	**List-II**
(A) Columnar epithelium	(I) Ducts of glands
(B) Ciliated epithelium	(II) Inner lining of stomach and intestine
(C) Squamous epithelium	(III) Inner lining of bronchioles
(D) Cuboidal epithelium	(IV) Endothelium

Choose the **correct** answer from the options given below :
(1) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(2) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

199. Match **List-I** with **List-II**.

List-I	**List-II**
(A) Cytokine barriers	(I) Mucus coating of respiratory tract
(B) Cellular barriers	(II) Interferons
(C) Physiological barriers	(III) Neutrophils and Macrophages
(D) Physical barriers	(IV) Tears and Saliva

Choose the **correct** answer from the options given below :
(1) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(3) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
(4) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

200. Select the **correct** statement/s with respect to mechanism of sex determination in Grasshopper.
(A) It is an example of female heterogamety.
(B) Male produces two different types of gametes either with or without X chromosome.
(C) Total number of chromosomes (autosomes and sex chromosomes) is same in both males and females.
(D) All eggs bear an additional X chromosome besides the autosomes.
Choose the **correct** answer from the options given below :
(1) (B) and (D) only
(2) (A), (C) and (D) only
(3) (A) only
(4) (A) and (C) only

ANSWER KEY NEET(UG)-2023(Manipur)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**1**	**3**	**4**	**4**	**3**	**2**	**4**	**1**	**4**	**4**	**4**	**4**	**4**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**4**	**3**	**4**	**2**	**4**	**3**	**3**	**1**	**4**	**1**	**1**	**3**	**4**	**4**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**1**	**2**	**2**	**4**	**1**	**3**	**2**	**3**	**3**	**2**	**4**	**2**	**2**	**4**	**1**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**1**	**3**	**1**	**2**	**3**	**2**	**3**	**2**	**3**	**4**	**4**	**2**	**4**	**4**	**4**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**1**	**1**	**3**	**4**	**3**	**4**	**1**	**1**	**3**	**3**	**1**	**1**	**3**	**4**	**3**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**1**	**3**	**4**	**4**	**2**	**3**	**3**	**4**	**4**	**4**	**4**	**4**	**4**	**3**	**3**
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	**4**	**2**	**2**	**3**	**3**	**2**	**1**	**1**	**1**	**1**	**3**	**4**	**1**	**1**	**1**
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	**1**	**4**	**2**	**4**	**4**	**3**	**1**	**4**	**4**	**1**	**1**	**1**	**1**	**2**	**1**
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	**4**	**4**	**4**	**1**	**1**	**3**	**2**	**1**	**4**	**2**	**3**	**2**	**2**	**1**	**1**
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	**2**	**2**	**3**	**4**	**1**	**1**	**2**	**3**	**3**	**2**	**4**	**4**	**3**	**4**	**2**
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	**2**	**3**	**4**	**3**	**4**	**1**	**4**	**3**	**3**	**2**	**2**	**1**	**1**	**4**	**4**
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	**3**	**1**	**4**	**2**	**4**	**3**	**3**	**2**	**2**	**1**	**2**	**4**	**4**	**1**	**3**
Que.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Ans.	**4**	**3**	**3**	**1**	**2**	**2**	**3**	**3**	**4**	**1**	**2**	**1**	**2**	**3**	**2**
Que.	196	197	198	199	200										
Ans.	**1**	**4**	**4**	**1**	**1**										

HINT - SHEET

1. By magnetic property

$$\chi \propto \frac{1}{T}$$

χ vs $\frac{1}{T}$ graph will be straight line.

2. In elastic collision maximum energy is transfer when M = m

3. $E_n = -13.6\frac{Z^2}{n^2}$

for ground state n = 1

[for second excited state n = 3]

$$\frac{E_3}{E_1} = \left(\frac{n_1}{n_3}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

$$E_3 = \frac{E_1}{9} = \frac{13.6}{9} = 1.51\text{eV}$$

4. $\boxed{V_\infty = \sqrt{V^2 - V_e^2}}$

Given than

$V = 2V_e$

So,

$$V_\infty = \sqrt{(2V_e)^2 - V_e^2}$$

$$V_\infty = \sqrt{3}V_e = 11.2\sqrt{3}\ \text{km/s}$$

5. Since lens is made of three materials so three different values of μ and hence three images.

6. Mean diameter $= \frac{d_1 + d_2 + d_3 + d_4 + d_5}{5}$

$$= \frac{3.33 + 3.32 + 3.34 + 3.33 + 3.32}{5}$$

$= 3.328 \approx 3.33$

7. Silver is good conductor so its resistivity will be very less.

8. Angular momentum = $[ML^2T^{-1}]$

Coeff of thermal conductivity = $[MLT^{-3}K^{-1}]$

Torque = $[ML^2T^{-2}]$

Gravitational constant = $[M^{-1}L^3T^{-2}]$

So, gravitational constant has power of (–1) of M.

9. $\vec{V} = \frac{d\vec{r}}{dt} = 4\hat{i} + 4t\hat{j} + 0\hat{k}$

at t = 1sec

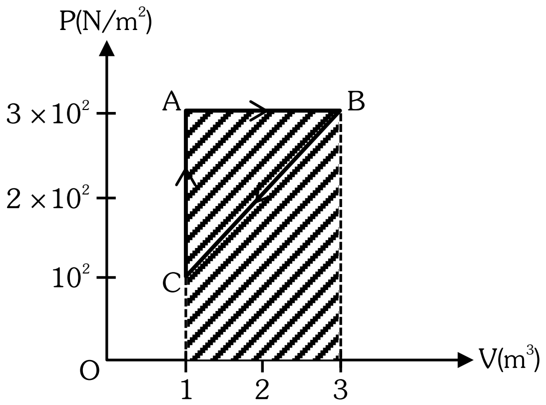
$\vec{V} = 4\hat{i} + 4(1)\hat{j}$

$|\vec{V}| = \sqrt{4^2 + 4^2} = 4\sqrt{2}$

$\tan\alpha = \frac{4}{4} = 1$

$\alpha = 45°$

10.



AB is isobaric process

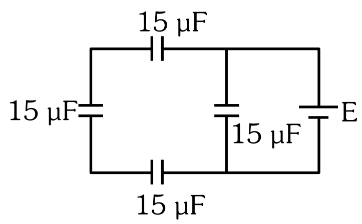
$W_{AB} = P(V_2 - V_1)$

$W_{AB} = 3 \times 10^2 (3-1)$

$W_{AB} = 3 \times 100 \times 2$

$W_{AB} = 600$ J

11.



$C_{eq} = 5 + 15 = 20\mu F$

12. $\tan\phi = \frac{X_L}{R}$

$X_L = \omega l = 100\pi \times \frac{1}{\pi} = 100\Omega$

$R = 100\Omega$

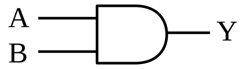
$\tan\phi = \frac{100}{100} = 1$

$\phi = 45°$

13. $Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}}.\overline{\overline{B}} = A.B$

AND gate

for AND gate



14. $P = \vec{F}.\vec{V}$

$P = (10\hat{i} + 10\hat{j} + 20\hat{k}).(5\hat{i} - 3\hat{j} + 6\hat{k})$

P = 50 – 30 + 120

$\boxed{P = 140 \text{ W}}$

15. $R = \frac{\rho L}{A}$ If diameter becomes thrice then cross

Section area will become 9 times so $R \propto \frac{1}{A}$

Resistance will become $\frac{1}{9}$ times

$R' = \frac{81\Omega}{9} = 9\Omega$

16. $\mu = \frac{C}{v} = \frac{\frac{1}{\sqrt{\mu_0 \epsilon_0}}}{\frac{1}{\sqrt{1.5\mu_0 \times 2\epsilon_0}}} = \sqrt{3}$

17. $\frac{\text{E.P.E}}{\text{Volume}} = \frac{1}{2}(\text{stress})(\text{strain})$

$= \frac{1}{2}(Y)(\text{strain})^2$

$= \frac{1}{2}(Y)\left(\frac{\Delta L}{L}\right)^2$

$= \frac{1}{2}(2 \times 10^{11})\left(\frac{1 \times 10^{-3}}{100 \times 10^{-2}}\right)^2$

$= 10^5$

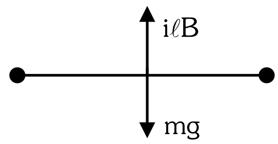
18. $n_{cop} = (2M+1)^{th}$ Har. $= (2 \times 4 + 1) \times \frac{V}{4\ell_c} = \frac{9V}{4\ell_c}$

$n_{oop} = (M+1)^{th}$ Har. $= (3 + 1)\ \frac{V}{2\ell_0} = \frac{4V}{2\ell_0}$

now $\frac{9V}{4\ell_c} = \frac{4V}{2\ell_0}$

$\frac{\ell_c}{\ell_0} = \frac{18}{16} = \frac{9}{8}$

19.



$mg = i\ell B$

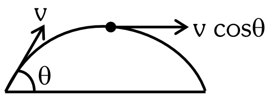
$250 \times 10^{-3} \times 9.8 = i \times 2 \times 0.7$

$i = \frac{0.250 \times 9.8}{2 \times 0.7} = 0.250 \times 7$

i = 1.75 A

20. $\phi = \frac{q_{inside}}{\varepsilon_0}$

only depends on charge enclosed by surface.

21. 

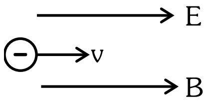
At the top most point of its trajectory particle will have only horizontal component of velocity

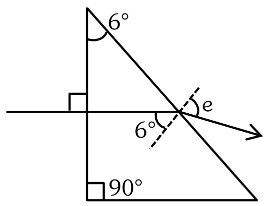
V at top = v cos θ

$= 20 \times \frac{1}{2} = 10$ m/s

22. Pressure is a scalar quantity.

23. Speed of electron will decrease due to electric force magnetic force of electron is zero.



24. 

$A = r_1 + r_2 = 6°$

$\mu = 1.5 \ \frac{\sin e}{\sin r_2} = \frac{\sin e}{\sin 6}$

$\sin e = 1.5 \times 6°$

$e \simeq 9°$

25. For p type semiconductor trivalent impurity is added

26. $\text{Sin } i_c = \frac{1}{\mu} \propto \lambda$

$i_c \propto \lambda$

Yellow, orange, red emerge from air.

27. $Z_1 = \sqrt{X_L^2 + R^2}$ $\qquad X_L = O$ (D.C. circuit)

$Z_1 = 10\Omega$

$Z_2 = \sqrt{X_C^2 + R^2}$

$X_C = \frac{1}{2\pi \times 50 \times \frac{10^3}{\pi} \times 10^{-6}} = 10\Omega$

$Z_2 = \sqrt{(10)^2 + (10)^2} = 10\sqrt{2}$

$Z_1 < Z_2$

28. $\frac{1}{\lambda} = R\left(\frac{1}{(1)^2} - \frac{1}{n^2}\right)$ $\quad n = 2, 3, 4, ...$

$\left(\frac{1}{\lambda_L}\right)_{max} = R\left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right)$ $\quad \left(\because \frac{1}{R} \simeq 912Å\right)$

$(\lambda_L)_{max} = \frac{4}{3}\frac{L}{R}$

$(\lambda_L)_{max} = \frac{4}{3} \times 912Å = 4 \times 304\ Å = 1216Å$

$\left(\frac{1}{\lambda_L}\right)_{min} = R\left(\frac{1}{(1)^2} - \frac{1}{(\infty)^2}\right)$

$(\lambda_L)_{min} = \frac{1}{R} \simeq 912Å$

Range of λ is 912Å to 1216Å which lies in U.V. region.

29. In given symbol, emitter current leave from emitter so transistor is NPN

order of doping E > C >> B

order of size C > E >> B

for active mode emitter base junction is forward bias and base-collector junction is reverse bias.

30. $\lambda_e = \frac{12.27}{\sqrt{V}}Å = \frac{12.27}{\sqrt{81}}Å = \frac{12.27}{9}Å$

$= 1.36\ Å \left(\because 1Å = \frac{1}{10}\text{nm}\right)$

$= 0.136$ nm

31. Power dissipated is maximum of purely resistive circuit.

32. Maximum kinetic energy of emitted electron is independent of intensity of radiation.

33. Final velocity of centre of mass = Initial velocity of centre of mass = 0 because net external force on system is zero.

34. Total flux from cube = $\frac{q}{\varepsilon_0}$

∴ So flux of any one surface of cube

$= \frac{q}{6\varepsilon_0} = \frac{Q \times 10^{-6}}{6\varepsilon_0}$

35. $F = 6\pi r\eta v$

$= (6)(3.14)\left(\frac{1 \times 10^{-3}}{2}\right)(0.8 \times 10^{-1})(2)$

$= 1.5 \times 10^{-3}$ N

36. $g = \frac{4}{3}\pi GR\rho$

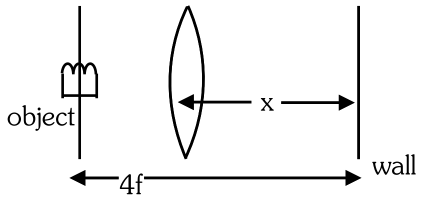
$\rho = \frac{3g}{4\pi GR}$

37. $I = neAV_d$

$V_d = \frac{I}{neA} = \frac{10}{10^{22} \times 1.6 \times 10^{-19} \times \pi \times 10^{-6}}$

$V_d = \frac{6.25}{\pi} \times 10^3$ m / sec

38.



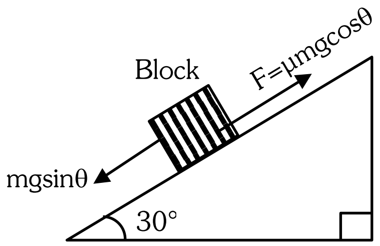
x = 2f
f = x/2

39. $\therefore V = \frac{KQ}{R}$

$E = \frac{KQ}{r^2}$

$E = \frac{VR}{r^2}$

40.



$\mu = \frac{1}{\sqrt{3}}$

$\tan 30^\circ = \frac{1}{\sqrt{3}}$

as $\mu = \tan\theta$

the block is at rest and net force on it must be zero

41. $P_{mix} = \frac{(\mu_1 + \mu_2)RT_{mix}}{V_{mix}}$

$= \frac{(0.2+0.3) \times 8.3 \times 200}{200 \times 10^{-6}}$

$= \frac{0.5 \times 8.3 \times 200}{200 \times 10^{-6}}$

$P_{mix} = 4.15 \times 10^6\ P_a$

42. Q = CV

$\frac{dQ}{dt} = C.\frac{dV}{dt} \Rightarrow \frac{dV}{dt} = \frac{I}{C} = \frac{2 \times 10^{-3}}{4 \times 10^{-6}}$

$= \frac{10^3}{2} = 500\frac{V}{s}$

43. $\omega = 2\pi \frac{rad}{sec}$

$E_{max} = NBA\omega$

$= 100 \times 0.05 \times 1 \times 2\pi$

$= 10 \times \pi$

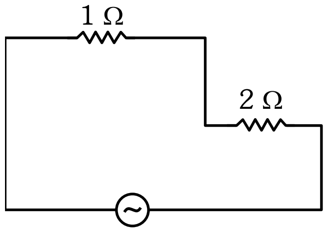
= 31.4 V

44. as frequency is very high

$X_C \approx 0$

$X_L \to \alpha$

Effective circuit will be



Effective impedance of circuit will be = 3Ω

45. $\tau = I\alpha \Rightarrow \alpha = \frac{\tau}{I} = \frac{100}{300} = \frac{1}{3}$ rad/sec^2

$\omega_i = 0$

$\omega_f = \omega_i + \alpha t$

$= 0 + \frac{1}{3} \times 3$

$\omega_f = 1$ rad/sec

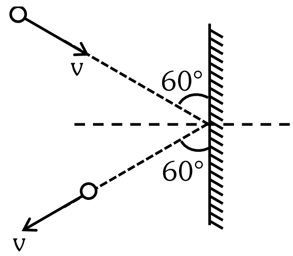
46. $r = \left(\frac{\ell_o - \ell_c}{\ell_c}\right)R$

$1 = \left(\frac{330 - \ell_c}{\ell_c}\right) \times 2$

$3\ell_c = 660$

$\ell_c = 220$ cm

47.



$F = \left|\frac{\Delta\vec{p}}{\Delta t}\right| = \frac{2mv\sin\theta}{t} = \frac{2(1)(1)\sin 60^\circ}{0.1} = 10\sqrt{3}\,N$

48. Given $mvr = 1.5\,\dfrac{h}{\pi}$

Compare with $mvr = n\dfrac{h}{2\pi}$

So $\dfrac{n}{2} = 1.5$ or $n = 3$

Now $E_3 = -\dfrac{13.6}{(3)^2}\,eV \simeq -1.5\,eV$

49. Direction of velocity and centripetal acceleration changes continuously so $\vec{v}$ is not constant and $\vec{a}$ is not a constant.

50. $T_{air} = 2\pi\sqrt{\dfrac{\ell}{g}} = \sqrt{3}$ sec

In Liquid $\rightarrow$ $g_{net} = g\left(1 - \dfrac{\rho}{\sigma}\right)$

ρ = density of liquid

σ = density of material of bob

so $T_{Liq} = 2\pi\sqrt{\left(\dfrac{\ell}{g_{net}}\right)} = 2\pi\sqrt{\dfrac{\ell}{g\left(1-\dfrac{\rho}{\sigma}\right)}}$

$$T_{Liq} = \frac{T_{air}}{\sqrt{1-\dfrac{\rho}{\sigma}}} = \frac{\sqrt{3}}{\sqrt{1-\dfrac{1}{4}}} = \frac{\sqrt{3}}{\dfrac{\sqrt{3}}{2}} = 2\,\text{sec}$$

51. $n = 5$, $\ell = 2$, $m = -2, -1, +1, +2$, $m_s = +\dfrac{1}{2}$

52. Reason is the correct explanation of Assertion.

53. Li, Be forms predominately covalent compounds.

54. $K_a = C\alpha^2$

$K_a = (0.1) \times (0.01)^2$

$K_a = 1 \times 10^{-5}$

55. $m = \dfrac{1000 \times M}{1000 \times d - MM_w}$

$m = \dfrac{1000 \times 1}{1000 \times 1.25 - 1 \times 85}$

$m = \dfrac{1000}{1165} = 0.858$

56.

$$CH_3\text{–}\underset{\underset{O}{\|}}{C}\text{–}CH_3 \xrightarrow[\text{aldol Rxn}]{Ba(OH)_2} CH_3\text{–}\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}\text{–}CH_2\text{–}\underset{\underset{O}{\|}}{C}\text{–}CH_3$$

Functional groups present in product are alcohol and ketone.

57. Tl^{+3} act as an oxidising agent not reducing agent.

58. $HF > NH_3 > H_2S > CH_4$

(Non-polar)

59. B and D statement are correct.

60. Cheilosis (Fissuring at corners of mouth and lips) occurs due to deficiency of vitamin B_2 (Riboflavin)

61. $E^{o}_{cell} = E^{o}_{C} - E^{o}_{A}$

$= (1.33) - (-0.44)$

$= +1.77$ V

62. $R\text{–}COOH \xrightarrow[\text{(ii) } H_2O/HCl]{\text{(i) } B_2H_6} R\text{–}CH_2OH$

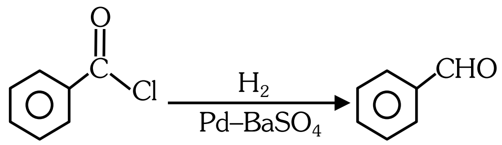
$R\text{–}CH{=}CH_2 \xrightarrow[\text{(ii) } H_2O, NaOH, H_2O_2]{\text{(i) } B_2H_6} R\text{–}CH_2\text{–}CH_2\text{–}OH$

63. $3A \rightarrow 2B$

$r = -\dfrac{1}{3}\dfrac{\Delta[A]}{\Delta t} = +\dfrac{1}{2}\dfrac{\Delta[B]}{\Delta t}$

$+\dfrac{\Delta[B]}{\Delta t} = -\dfrac{2}{3}\dfrac{\Delta[A]}{\Delta t}$

64. Rosenmund reaction



65. Equanil is used in controlling depression and hyper tension.

66. Total numbers electrons are same

Ca^{+2}, Ar, K^+, Cl^- $\rightarrow$ 20 electrons

67. Statement-I is true and statement-II is false.

68. $\left(P + \dfrac{a}{V^2}\right)(V - b) = RT$

69. $\rho = \dfrac{PM}{RT}$

For maximum density, pressure should be maximum and temperature should be minimum.

70. Mn_2O_7, SO_2, TeO_3 are acidic oxides.

71. Methylene blue solution

72. (A) Glycerol from spent lye $\rightarrow$ Distillation under reduced pressure

(B) Chloroform + Aniline $\rightarrow$ Distillation

(C) Fractions of crude oil $\rightarrow$ Fractional distillation

(D) Aniline + H_2O $\rightarrow$ Steam distillation

73. [B] R–X + Ag–C≡N $\rightarrow$ R–NC

Isonitrile

74. $LaH_{2.87}$ $\rightarrow$ non-stoichiometric

$\rightarrow$ Metallic/ Interstitial hydride.

75. In the formation of BMO, the two electron waves of the bonding atoms reinforce each other due to constructive interference.
Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are 'unsymmetrical' around bond axis.

76. Acidic buffer is prepared by mixing weak acid and its salt with strong base.

77. $$R-CH_2-OH \xrightarrow[K_2Cr_2O_7-H_2SO_4 \text{ or}]{CrO_3-H_2SO_4 \text{ or}} R\text{–}COOH$$
$$KMnO_4 + KOH / H_3O^+$$
[Strong oxidising agents]

78. Maximum covalency of boron is four.

79. $r = k[A]^{-1/2}[B]^{3/2}$

$$\text{order} = -\frac{1}{2}+\frac{3}{2}=\frac{2}{2}=1$$

80.

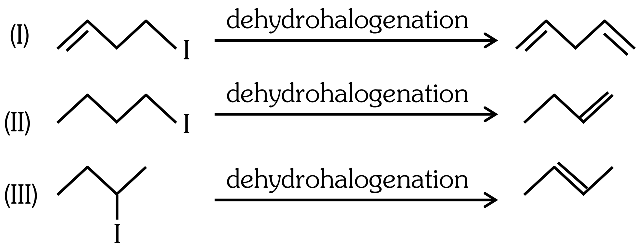
81. Complex salt is $K_2[Pt(NH_3)_2Cl_2]$
Double salt is $KAl(SO_4)_2.12H_2O$ (potash alum)

82. NCERT Pg.436 (Polymer)

83. $$CH_3-H_2C-\overset{sp^2}{\underset{CH_3}{C}}=\overset{sp^2}{\underset{CH_3}{C}}-CH_2-CH_3$$

84. Hydrated chlorides and Bromides of Ca, Sr and Ba are Ionic so undergo dehydration after heating.
Hydrated chlorides and Bromides of Be and Mg are covalent so undergo hydrolysis on Heating.

85.



Rate of Dehydrohalogenation : II < III < I

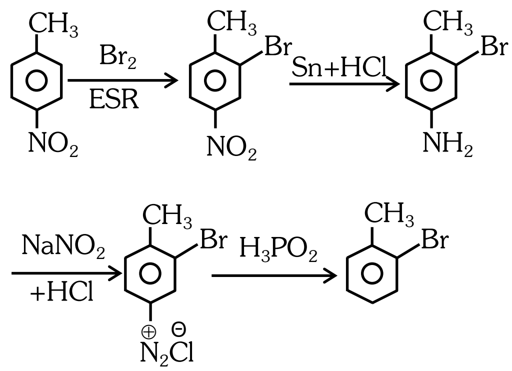
86. Number of particles = $5N_A$
Number of THV = 2 × number of particles, for close packing
$= 2 \times 5N_A$
$= 10\ N_A$
$= 10 \times 6.023 \times 10^{23}$
$= 6.023 \times 10^{24}$

87. Given complex compounds exhibit solvate isomerism having co-ordination number = 6.

88. $NO \rightarrow$ neutral $CaO \rightarrow$ Basic
$CO_2 \rightarrow$ Acidic $ZnO \rightarrow$ Amphoteric

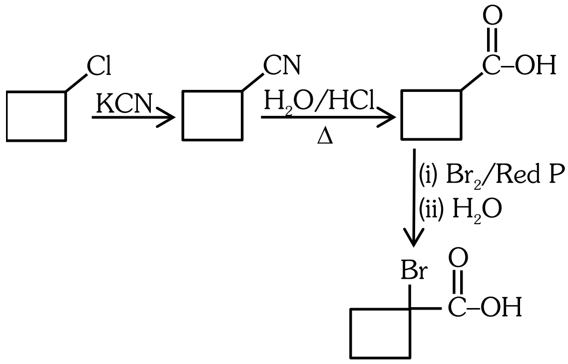
89. Uses in metallurgy (NCERT)

90.



91. For CCP (FCC)
$4r = \sqrt{2}a$
$a = \frac{4r}{\sqrt{2}}$
$a = 2\sqrt{2}r$

92.



93.

94. Redox couple is both the reduced and oxidised form involve same element.

95. Reason in correct explanation the above Assertion.

96. Decomposition for 1 mole of water

$$H_2O(g) \rightarrow H_2(g) + \frac{1}{2}O_2(g)\ ;\ \Delta H = +\frac{483.64}{2}$$

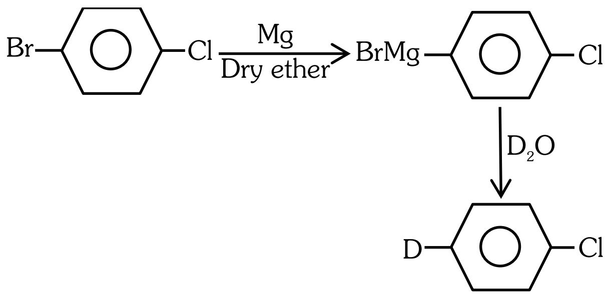
$\Delta H = +\ 241.82$ kJ

97.

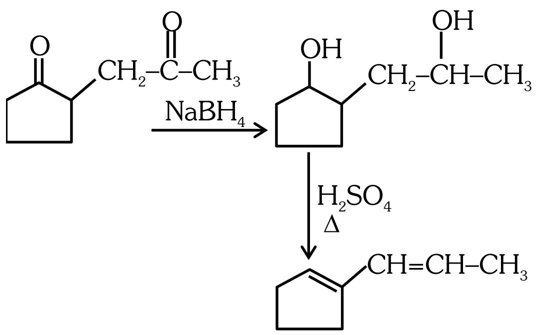
98. $i \times M \downarrow \Rightarrow \Delta T_b \downarrow$

Electrolyte	i ×M
NaCl	$2 \times 0.05 = 0.1$
KCl	$2 \times 0.1 = 0.2$
$MgSO_4$	$2 \times 0.1 = 0.2$
NaCl	$2 \times 1 = 2$

99.



100.



101. NCERT XI Pg # 144, 148, 149

102. NCERT XII Pg # 250,251

103. NCERT XI Pg # 74

104. NCERT XI Pg # 210

105. NCERT XII Pg # 276, 277

106. NCERT XII Pg # 74,76,80

107. NCERT XI Pg # 231

108. NCERT XI Pg # 168

109. NCERT XI Pg # 183

110. Module # 02

111. NCERT XI Pg # 218, 220

112. NCERT XII Pg # 263

113. NCERT XI Pg # 74

114. NCERT XII Pg # 24

115. NCERT XII Pg # 28

116. NCERT XI Pg # 247

117. NCERT XI Pg # 87,90,91

118. NCERT XII Pg # 275

119. NCERT XII Pg # 119

120. NCERT XII Pg # 253

121. NCERT XII Pg # 237

122. Module XII

123. NCERT XI Pg # 38

124. NCERT XI Pg # 148

125. NCERT XII Pg # 203

126. NCERT XII Pg # 117

127. NCERT XI Pg # 74

128. NCERT XI Pg # 235

129. NCERT XII Pg # 199

130. NCERT XI Pg # 248, 249, 250

131. NCERT XI Pg # 177

132. NCERT XI Pg # 33

133. NCERT XII Pg # 113

134. NCERT XII Pg # 234

135. NCERT XI Pg # 95

136. NCERT XI Pg # 198

137. NCERT XI Pg # 213

138. NCERT XI Pg # 201

139. NCERT XII Pg # 26,27

140. NCERT XI Pg # 42

141. NCERT XII Pg # 200

142. NCERT XI Pg # 85

143. NCERT XII Pg # 199

144. NCERT XII Pg # 107

145. NCERT XII Pg # 102
146. NCERT XII Pg # 74,76
147. NCERT XI Pg # 166, 169
148. NCERT XI Pg # 240
149. NCERT XI Pg # 72,73
150. NCERT XI Pg # 175, 190
151. NCERT XII Pg # 63
152. NCERT XII Pg # 147, 148, 149
153. NCERT XI Pg # 146
154. NCERT XII Pg # 47
155. NCERT XI Pg # 11
156. NCERT XII Pg # 211
157. NCERT XI Pg # 164
158. NCERT XII Pg # 103
159. NCERT XI Pg # 294
160. NCERT XI Pg # 114
161. NCERT XI Pg # 334, 335
162. NCERT XI Pg # 270
163. NCERT XII Pg # 264, 265
164. NCERT XI Pg # 21,49,51,54
165. NCERT XI Pg # 311
166. NCERT XII Pg # 60
167. NCERT XII Pg # 195
168. NCERT XII Pg # 113
169. NCERT XII Pg # 64
170. NCERT XI Pg # 286
171. NCERT XI Pg # 279, 280
172. NCERT XI Pg # 102
173. NCERT XII Pg # 48
174. NCERT XII Pg # 227
175. NCERT XI Pg # 323
176. NCERT XII Pg # 117
177. NCERT XII Pg # 276,281,283,284
178. NCERT XII Pg # 159
179. NCERT XI Pg # 6
180. NCERT XII Pg # 58,61
181. NCERT XI Pg # 146
182. NCERT XI Pg # 128, 129
183. NCERT XI Pg # 168
184. NCERT XI Pg # 321
185. NCERT XII Pg # 141
186. NCERT XII Pg # 102
187. NCERT XI Pg # 265
188. NCERT XII Pg # 226
189. NCERT XII Pg # 112
190. NCERT XI Pg # 297
191. NCERT XII Pg # 274,275
192. NCERT XI Pg # 335
193. NCERT XI Pg # 49,51
194. NCERT XII Pg # 209,211,212
195. NCERT XI Pg # 134
196. NCERT XII Pg # 168,169
197. NCERT XI Pg # 112
198. NCERT XI Pg # 101
199. NCERT XII Pg # 150,151
200. NCERT XII Pg # 85